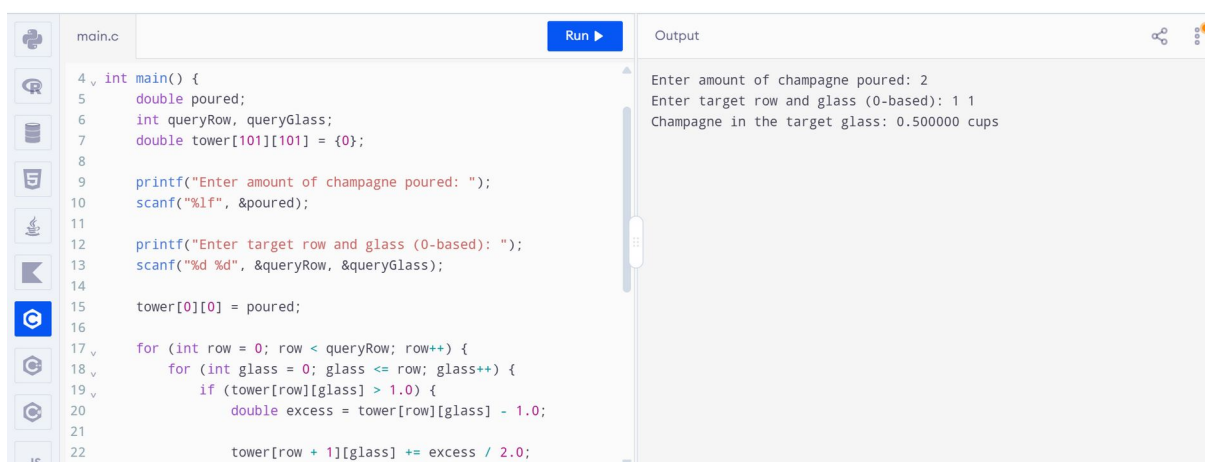


Aim

To determine how much champagne remains in a particular glass of a champagne tower after pouring a specified quantity into the top glass, using Dynamic Programming.

Algorithm

1. Read the amount of champagne poured, the target row, and the target glass.
2. Initialize the top glass with the poured quantity.
3. If a glass contains more than 1 cup, keep 1 cup in it.
4. Distribute half of its excess champagne to each of the two glasses directly below it.
5. Repeat the process row by row.
6. Display the amount of champagne in the target glass, limited to 1 cup



```
main.c Run ▶ Output
4 int main() {
5     double poured;
6     int queryRow, queryGlass;
7     double tower[101][101] = {0};
8
9     printf("Enter amount of champagne poured: ");
10    scanf("%lf", &poured);
11
12    printf("Enter target row and glass (0-based): ");
13    scanf("%d %d", &queryRow, &queryGlass);
14
15    tower[0][0] = poured;
16
17    for (int row = 0; row < queryRow; row++) {
18        for (int glass = 0; glass <= row; glass++) {
19            if (tower[row][glass] > 1.0) {
20                double excess = tower[row][glass] - 1.0;
21
22                tower[row + 1][glass] += excess / 2.0;
```

Enter amount of champagne poured: 2
Enter target row and glass (0-based): 1 1
Champagne in the target glass: 0.500000 cups

Explanation

Two cups are poured into the top glass.

- The top glass retains 1 cup.
- The remaining 1 cup overflows.
- Half of the excess goes to the left glass in row 1.
- The other half goes to the right glass in row 1.

Thus, the target glass at row 1, glass 1 contains 0.5 cups.

Complexity Analysis

- Time Complexity: $O(R^2)$, where R is the target row.

- Space Complexity: $O(R^2)O(R^2)O(R^2)$, because a two-dimensional array stores the champagne distribution.